

Emmanuel Idoko

emmanuel.idoko@gmail.com | linkedin.com/in/emmanuelidoko | github.com/pidoxy

EDUCATION

University of Lagos

Bachelor of Science in Computer Science | CGPA: 4.68/5.0

Lagos, Nigeria

Expected 2027

- **Relevant Coursework:** Discrete Mathematics, Data Structures, Computational Methods, Linear Algebra, Calculus, Python/C Programming, Algorithm and Complexity, Compiler Construction, Concurrent Programming, Operating Systems, Distributed Systems, Net Centric Computing, Systems Analysis and Design, Analysis and Design of Digital Systems, Operations Research Techniques, Machine Assembly Language.

African Computer Vision Summer School (ACVSS)

3rd ACVSS, Google AI Community Center

Accra, Ghana

Jul 2026

- **Sessions:** 10 lectures and 8 practical sessions covering geometry and mathematics for vision, deep learning and AI, and ethics in computer vision, alongside 8 keynotes, a hackathon, and a poster session.

Udacity

AI Programming with Python Nanodegree

Remote

Oct 2022 – Jan 2024

EXPERIENCE

AidCare

Founder & AI/ML Engineer

Lagos, Nigeria (Remote)

May 2025 – Present

- Designed and deployed a full-stack clinical AI platform on DigitalOcean with a containerized FastAPI backend, RBAC, scoped tokens, and anonymized logging — meeting data privacy requirements for healthcare deployment.
- Built a semantic search pipeline using Sentence Transformers and FAISS, indexing 500+ structured Nigerian clinical protocols for sub-second retrieval at inference time.
- Integrated Whisper ASR and Tesseract OCR into a multimodal ingestion pipeline, enabling voice-driven and document-based query workflows for low-resource clinical environments.
- Engineered a token-efficient LLM prompting layer (Google Gemini) for grounded differential diagnosis generation, reducing lookup time for healthcare staff by 25%.

HabariPay – GTCO

Software Engineer, Contract

Lagos, Nigeria (Hybrid)

Sep 2024 – Present

- Currently building enterprise events software for Squad, the fintech subsidiary arm of GTCO, supporting event management and payment workflows at scale.
- Built a full-stack vendor guide application (Next.js, Node.js, PostgreSQL) for a food festival, serving 260+ attendees and generating 2,100+ page views.
- Automated ingestion of vendor data from 2 sources into a centralized SQLite database, processing hundreds of records with high consistency.
- Implemented a TF-IDF recommendation engine and Gemini-powered summaries, improving user click-through rate by 7%.

Wema Bank

Data and AI Internship

Lagos, Nigeria (Hybrid)

Sep 2024 – Dec 2024

- Built a governance-compliant pipeline to extract and analyze 10,000+ Play Store reviews, producing structured product insights for cross-functional teams.
- Contributed to an AI-powered knowledge chatbot for Microsoft Teams (Node.js, Azure), enabling 2,000+ staff to query internal product and process documentation instantly.

NexaScale

Cloud Engineer Internship

Remote

Feb 2023 – May 2023

- Automated deployment of 30+ AWS resources with Terraform, reducing manual infrastructure setup by 75%.
- Built and deployed a microservice-based application with Node.js, containerized with Docker, and orchestrated on Kubernetes (AKS).
- Established a CI/CD pipeline using Azure DevOps, Git, and Docker for reliable, repeatable production deployments.

ZuriChat

Software Engineer & Lead Internship

Remote

Jul 2021 – Oct 2021

- Led a cross-functional team to ship real-time music and chess rooms using WebRTC, increasing user engagement by 20%.
- Improved backend performance by 50% through caching strategies, code refactoring, and API optimization.
- Managed multi-team deployments using GitHub Actions and Jenkins, ensuring stable integrations across concurrent feature branches.

PROJECTS

Virtue Foundation Intelligence Platform

Python, LangGraph, FAISS, GPT-4o-mini, FastAPI, Next.js

- Built an end-to-end agentic AI platform analyzing 797 Ghana healthcare facilities across 16 regions, identifying 10 medical deserts and 43 data anomalies through a multi-agent pipeline.
- Designed an Intelligent Document Processing engine (GPT-4o-mini) to extract structured capability data from unstructured facility records, with per-field confidence scoring (0-99%).
- Architected a LangGraph supervisor system routing natural language queries to 6 specialized sub-agents backed by a FAISS vector store on OpenAI embeddings; integrated ElevenLabs voice I/O for field accessibility.
- Won 2nd place at Databricks x Hack-Nation Global AI Hackathon.

Kinetix

Python, FastAPI, Next.js, TypeScript, Gemini API, WebSockets

- Built a real-time AI personal trainer using Gemini's vision API to analyze webcam exercise footage frame-by-frame, classifying posture and joint alignment as safe, suboptimal, or unsafe.
- Implemented WebSocket communication between FastAPI backend and Next.js frontend for low-latency feedback delivery; skeleton overlay with color-coded form indicators (green/yellow/red).
- Designed a dual-persona prompt architecture separating internal biomechanical reasoning from external coaching output, with post-session summaries and corrective exercise recommendations.

Modern Search Engine

C#, .NET, Azure, JavaScript

- Engineered a distributed information retrieval system with a custom inverted index, TF-IDF weighting, and cosine similarity, delivering sub-second query responses across 10,000+ PDF, DOCX, and TXT files.
- Designed and validated a robust text-processing pipeline (tokenization, stemming, stop-word removal) with comprehensive xUnit test coverage.

PheraCAM

Python, React.js, Azure, OpenAI API

- Developed a real-time face recognition system using OpenCV and Azure cloud services, achieving >95% accuracy with <150ms response time.
- Won 2nd place among 50+ teams at Hacklab's Hackathon, recognized for technical innovation and robustness.

AFCF Portal

Node.js, React.js

- Built a national reporting and coordination platform for 500+ users, automating compliance workflows and cutting manual data collation time by 75%.

Data Structures and Algorithms - WolfStation

Python, Tries, Algorithm Design

- Engineered a high-performance contact lookup system using an array-implemented trie, achieving a 20% performance gain and 98.73% space optimisation versus alternative implementations.
- Hosted and led problem-solving sessions on algorithm optimisation, solving complex challenges in Python.

RESEARCH & PUBLICATIONS

SharpXR: Structure-Aware Denoising for Pediatric Chest X-Rays

MIRASOL Workshop, MICCAI 2025 | arXiv:2508.08518

- Benchmarked seven denoising baselines; contributed to experimental evaluation and manuscript writing. SharpXR improved downstream pneumonia classification from 88.8% to 92.5%.

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, JavaScript, TypeScript, SQL, Bash
- **Frameworks:** Node.js, Express, Next.js, Flask, FastAPI, React, Pandas, NumPy, OpenCV, PyTorch
- **Systems & Tools:** Docker, Kubernetes, Terraform, AWS, Azure, DigitalOcean, Git, Linux, CI/CD (GitHub Actions, Jenkins, Azure DevOps)
- **AI & Data:** LangGraph, FAISS, Hugging Face Transformers, Sentence Transformers, Scikit-learn, Whisper, OpenAI API, Gemini API

SCHOLARSHIPS & AWARDS

- Jim Ovia Undergraduate Scholarship (2025-2027): National scholarship recognizing top performance in STEM.
- Squad Hackademy Undergraduate Scholarship (2024-2027): Recognizing top performance in programming.
- PICFI Undergraduate Scholarship (2025-2027): National scholarship recognizing top performance in STEM.
- Meta Global Hackathon (Jun 2022): Placed 14th in EMEA and 47th globally; engineered 10+ algorithm solutions to address critical system bugs, reducing error rates by 40% and improving stability by 25%.